

01

TCP vs UDP

Two transport protocols, both seen in one secure website visit

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Feature	TCP	UDP
Connection	Handshake first	None, sends at once
Reliability	ACKs + retransmission	No guarantee
Ordering	Sequence numbers	No ordering
Header / PDU	20+ bytes / segment	8 bytes / datagram
Speed	Slower, more overhead	Fast, low overhead
In my capture	HTTPS, port 443	DNS, port 53

TCP session to 172.66.0.227:443: SYN, SYN-ACK, ACK, then TLS Client Hello (SNI = x.com).



UDP: DNS query, source port 57452 to port 53, length 31

58.3% packets were TCP (HTTPS/TLS)

3.1% packets were UDP (DNS)

Application examples

The right protocol depends on what the application can tolerate

TCP

Reliability first

- **Secure web (HTTPS, 443)** every byte must arrive in order
- **Email (SMTP, IMAP)** messages must not be lost
- **File transfer (FTP, SFTP)** one missing block corrupts the file
- **Remote access (SSH, RDP)** commands must be exact

UDP

Speed first

- **DNS lookups (53)** one small question, one small answer
- **Video meetings, VoIP** a late packet is useless
- **Live streaming, gaming** current data beats retransmission
- **DHCP, NTP** short request and reply

Communication failure

DNS not responding: website cannot be opened, network is healthy

What happened

- **Symptom:** "Could not resolve host: x.com"
- **Cause:** DNS server set to 192.0.2.1, where nothing answers
- **Layer:** Application (DNS over UDP port 53)
- **Key clue:** ping 8.8.8.8 still works, so IP and routing are fine

Error seen by the user

```
C:\Windows\System32>ping x.com -n 2
Ping request could not find host x.com. Please check the name and try again.

C:\Windows\System32>curl.exe -I https://x.com
curl: (6) Could not resolve host: x.com
```

Wireshark: repeated DNS queries to 192.0.2.1, no reply

No.	Time	Source	Destination	Protocol	Length	Info
17	25.794000	192.168.1.10	192.0.2.1	DNS	65	Standard query 60201 HTTP x.com
18	25.794463	192.168.1.10	192.0.2.1	DNS	65	Standard query 60201 AAAA x.com
19	25.791181	192.168.1.10	192.0.2.1	DNS	65	Standard query 60200 A x.com
22	26.816660	192.168.1.10	192.0.2.1	DNS	65	Standard query 60202 HTTP x.com
23	26.816993	192.168.1.10	192.0.2.1	DNS	65	Standard query 60202 AAAA x.com
24	26.811282	192.168.1.10	192.0.2.1	DNS	65	Standard query 60279 A x.com
25	27.918612	192.168.1.10	192.0.2.1	DNS	70	Standard query 60279 HTTP gator.voices.com
26	27.918783	192.168.1.10	192.0.2.1	DNS	70	Standard query 60262 AAAA gator.voices.com
27	27.918917	192.168.1.10	192.0.2.1	DNS	70	Standard query 60262 A gator.voices.com

Second test, port 444 blocked: SYN sent and retransmitted 4 times, never answered. A firewall is likely dropping it.

Troubleshooting procedure

Bottom-up: test one layer at a time, then fix and verify

1 Record the symptom

Website fails with "Could not resolve host"

2 ping 8.8.8.8

Replies: internet and routing work

3 nslookup x.com

Fails: name resolution is the fault

4 ipconfig /all

DNS server is 192.0.2.1, not the router

5 Wireshark capture

DNS queries sent, no replies

6 Fix and verify

Reset DNS to 192.168.1.1, flush cache, retest

Verified: gateway ping, 8.8.8.8 ping, nslookup, port 443 test, tracert (8 hops), curl HTTP 200 OK